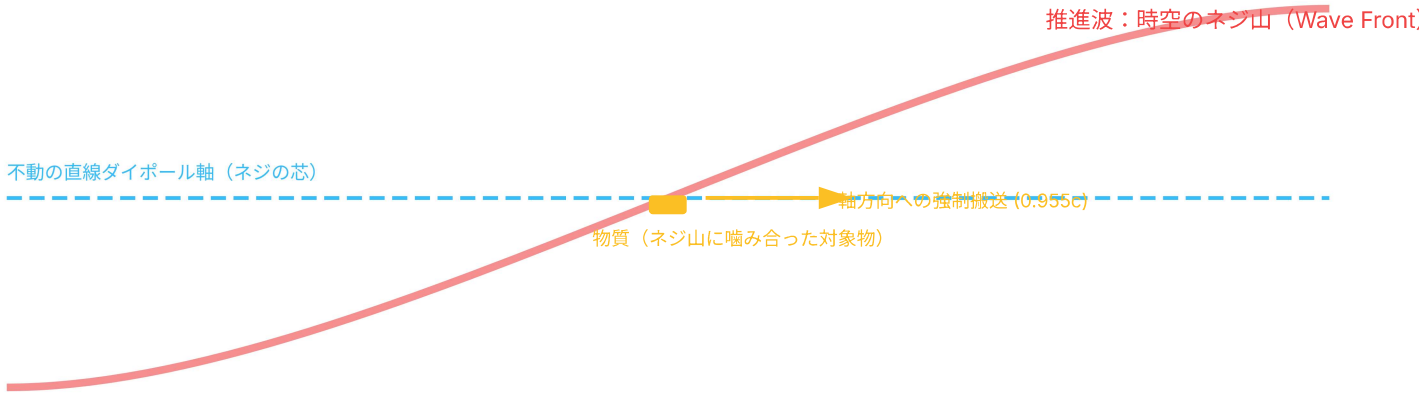


## Spacetime screw effect: propulsion wave structure on a linear axis



\*The shaft does not rotate; the screw threads (red line) slide linearly, physically pushing the material upward.

## Dynamic interference between "screw threads" on a linear axis and material

### Section 2: Generation of phase lag due to thread pitch compression

The "phase lag" (mismatch of peaks) observed in quasar distributions is caused by a localized increase in the pitch density of the propulsion waves (screw threads) traveling through spacetime.

**Physical mechanism:** When the spacing between wave faces (equal phase surfaces) incident at an oblique angle to the linear axis narrows, the efficiency of propulsion force transmission from the wave to the material changes, and geometric "slip" accumulates. This is the phase shift in the observed data.  $\Delta\phi$  This is its true identity.

This phenomenon is particularly pronounced at the "nodes" of spacetime aligned along a spiral spatial structure, causing a relative shift in the trapping position of matter.

## Mechanical order of spacetime according to the linear axis model

### Section 3: The "screw thread" structure of spacetime that transmits power

#### Unmoving core (dipole axis)

Non-rotating linearity ensures coordinate stability and the directionality of bulk flow across the entire universe.

#### Equiphasic surfaces of propulsion waves

A "wave wall" moves along the axis. These threads physically grip the material, driving a flow of 0.955c.

"The universe is an intricate mechanical system that transports matter using straight cores and sliding screw threads."

# WRM Technical Appendix

A physical analysis based on the "screw threads" and "surfboard effect" of spacetime.

## 1. The Screw Threads of Spacetime (The Geometry of Propulsion Waves)

The universe's main axis (dipole axis) is a straight line that does not rotate, but the standing waves that travel along it have a geometric structure like a "screw thread".

### The role of propulsion waves (equal phase surfaces)

When the wavefront propagates at an angle rather than perpendicular to the axis, a "physical upward force" is generated in the axial direction. This becomes a carrier wave on a cosmic scale.

### Material trap

The substance is the object that engages with the "grooves" of this screw thread. As the substance moves through space (0.955c), it is forcibly transported axially.

## 2. Surfboard effect and slip rate $\alpha$

Why do the observational data contain tiny fractional values? It's because the material is like a "surfboard" riding a wave, and "slipping" occurs, meaning it cannot perfectly synchronize with the wave.

$$\alpha = 0.046 \quad (\approx 18.2^\circ)$$

This value is the "pitch angle of the screw thread," and it represents the geometric ratio of how much material slides down (or encounters resistance to) the slope of the wave while traveling one unit along the axis.

This slip is the true nature of the "unexplained deviation" that appears on our instruments as the residual orbital difference or Hubble tension of the DE440.

## 3. Accumulation of phase lag and pitch compression

The "mountain discrepancy" observed in distant observations of quasars and other geometrid objects is the sum of the slip accumulated during the process of passing through multiple turns of the screw.

- Pitch reduction:** When the spacing between threads narrows near a node, the number of slips per unit distance increases.
- Phase lag  $\Delta\phi$**  The further away the object is, the more frequent the "number of contacts with the screw threads" becomes, resulting in a more pronounced discrepancy with localized observations.

### Implications for self-cooling mechanisms

It is discharged as a slip component. 0.046c This energy may be acting as an "escape route for entropy" that prevents heat buildup within the system.  
The universe is a giant "standing wave screw" that carries information while cooling itself.